MLFlow

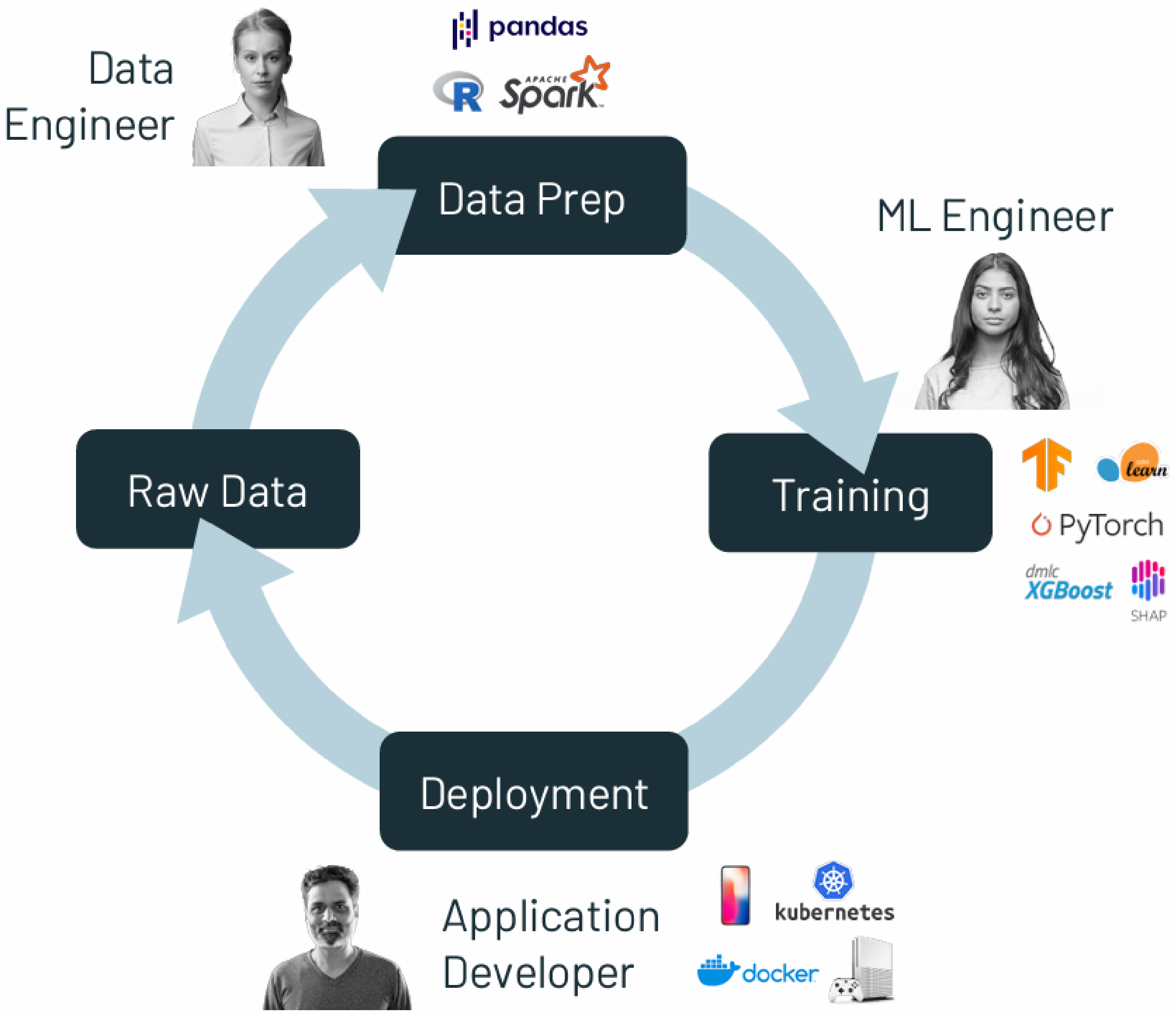
Track, Log, Load, Compare and Deploy ML Models

Outline

- Understand what MLflow is and why we use it.
- Track experiments and hyperparameters.
- Log metrics, models, and artifacts.
- Load and compare previous model runs.
- Serve and deploy ML models with MLflow.

Production ML is Even Harder

- ML apps must be fed new data to keep working.
- Design, retraining & inference often done by different people



A Solution is Emerging: ML Platforms

Software to manage the ML development and deployment process, from data to experimentation to production

Examples: Google TFX, Facebook FBLearner, Uber Michelangelo

Typical concerns:

- Data management
- Model management
- Reproducibility
- Experiment management
- Deployment for inference
- Testing & monitoring

What is MLflow

MLflow is an open-source platform to manage the ML lifecycle, including:

- Experiment Tracking – Record parameters, metrics, and results.
- Model Registry – Store, version, and manage trained models.
- Project Packaging – Define reproducible runs (like Docker or Conda).
- Model Serving – Deploy trained models easily with one command.

MLflow

Based on an open interface design philosophy: make it easy to connect arbitrary ML code & tools into the platform

Simple command-line and REST APIs rather than environment-specific

mlflow TRACKING
Experiment and metric tracking

mlflow PROJECTS
Reproducible execution

mlflow MODELS
Model packaging and deployment

mlflow MODEL REGISTRY
Model management

TensorFlow, PyTorch, Keras, Spark, scikit-learn, RAPIDS, H2O.ai, python, R, Java, mleap, ONNX, GLUON, XGBoost, LightGBM, spaCy, fast.ai, CONDA, kubernetes, docker, Amazon SageMaker, Azure Machine Learning, Google Cloud, databricks

Built-in Connectors

MLFlow Tracking

MLflow Tracking provides experiment logging, parameter tracking, metrics visualization, and model artifact management which are explained below:

- Experiment Organization: Track and compare multiple model experiments
- Metric Visualization: Built-in plots and charts for model performance
- Artifact Storage: Store models, plots, and other files with each run
- Collaboration: Share experiments and results across teams

MLFlow Tracking

Model Registry

MLflow Model Registry provides centralized model versioning, stage management, and model lineage tracking.

- Version Control: Track model versions with automatic lineage (model version with parameters and data)
- Stage Management: Promote models through staging, production, and archived stages
- Collaboration: Team-based model review and approval workflows (Offers REST APIs with model)
- Model Discovery: Search and discover models across your organization (Model Resusability)

Model Registry

mlflow 2.10.0 Experiments Models GitHub Docs

Registered Models

Create Model

Filter registered models by name o...

Name	Latest version	Aliased versions	Created by	Last modified	Tags
iris_model_dev	Version 17			2023-09-25 12:50:...	—
iris_model_prod	Version 11	@ champion : Version 11 +3		2023-10-26 17:10:...	—
iris_model_staging	Version 11			2023-09-25 12:46:...	—
iris_model_testing	Version 1			2023-09-27 13:17:...	—
mnist_model_dev	Version 12			2023-09-25 12:39:...	—
mnist_model_prod	Version 8	@ challenger : Version 8 +1		2024-01-19 10:35:...	—
mnist_model_staging	Version 8			2023-09-25 12:51:...	—

New model registry UI

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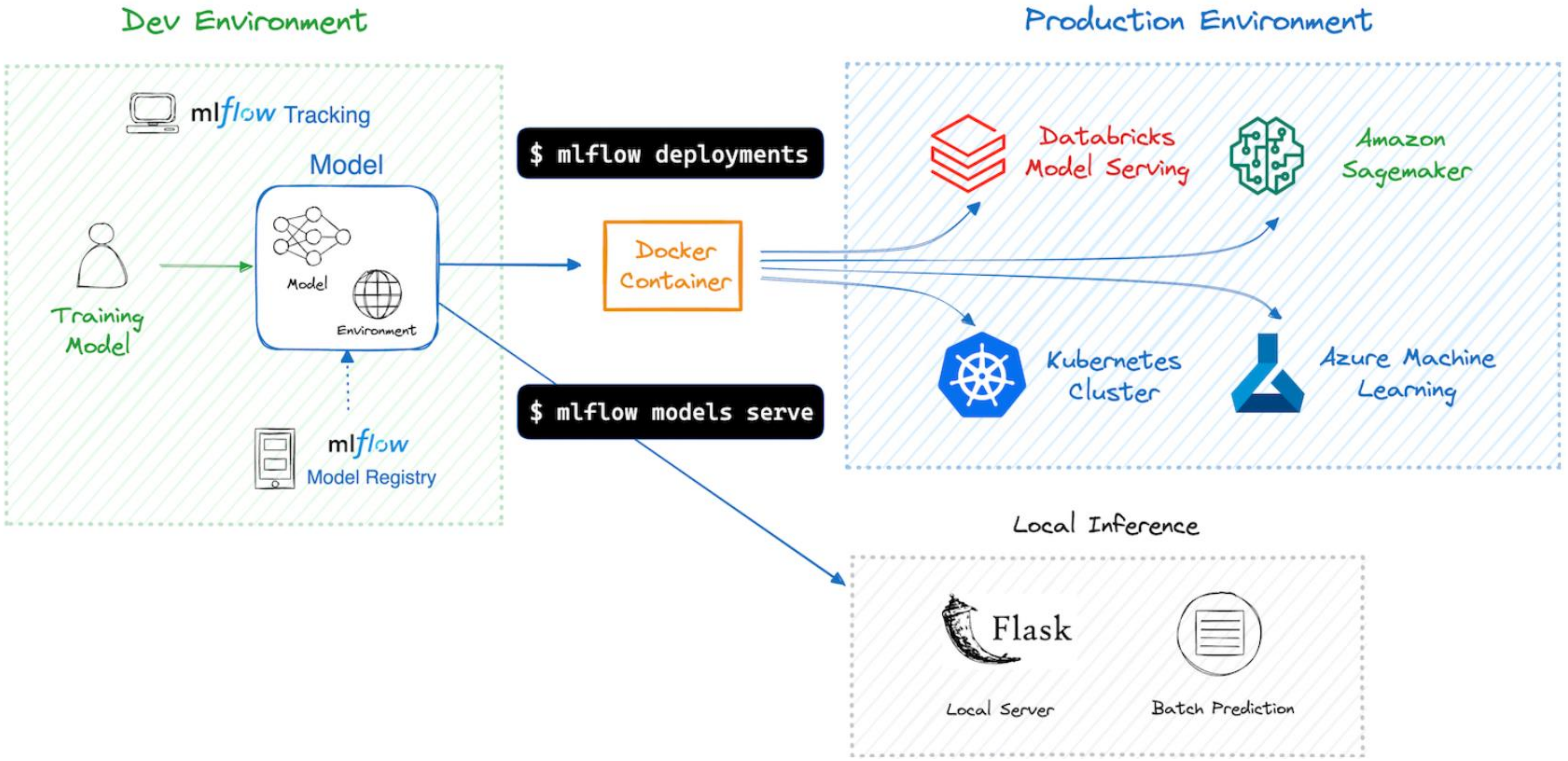
Model Deployment

MLflow Deployment supports multiple deployment targets including REST APIs, cloud platforms, and edge devices, which include:

- Multiple Targets: Deploy to local servers, cloud platforms, or containerized
- Model Serving: Built-in REST API serving with automatic input validation
- Batch Inference: Support for batch scoring and offline predictions
- Production Ready: Scalable deployment options for enterprise use

Model Deployment

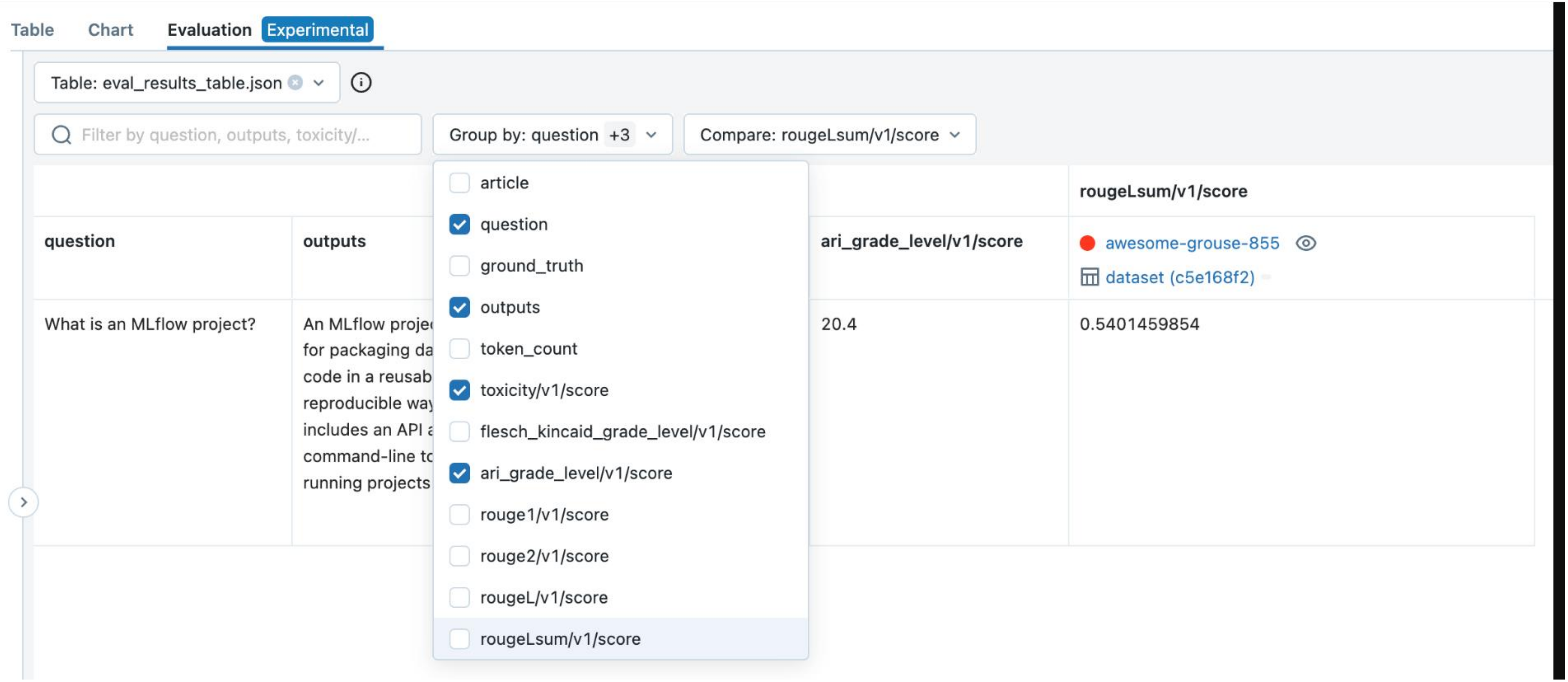


Model Evaluation

MLflow Evaluation provides comprehensive model validation tools, automated metrics calculation, and model comparison capabilities which are as follows:

- Automated Metrics: Built-in evaluation metrics for classification, regression, and - mo
- Custom Evaluators: Create custom evaluation functions for domain-specific metrics
- Model Comparison: Compare multiple models and versions side-by-side
- Validation Datasets: Track evaluation datasets and ensure reproducible results

Model Evaluation



Class Exercise:

- Understanding MLFlow and UI
- Build a machine learning model and interface hyper-parameters on MLFlow.
- Log model artifacts.
- Comparative analysis b/w different models.
- Register a best performing model to Registry.
- Interface model via REST API.

REST API

- Serve the model via REST API:

```
mlflow models serve -m "runs:/<RUN_ID>/model" -p 1234
```